# No. of Functional Features Included in the Solution

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| **Date** | 3 October 2026 |
| **Team ID** | SWTID-2026-9282 |
| **Project Name** | ComicCraft-AI Comic Creator using Gemini Models |
| **Maximum Marks** | 5 Marks |

### Functional Features Overview:

This document lists the functional features included in the **ComicCraft – AI Comic Story Creator using Gemini Models** project. Each feature describes its purpose, implementation status and contribution to the overall solution.

| S.No | Feature Name | Feature Description | Module / Component | Status (Done / In Progress / Pending) | Marks Contribution |
|---|---|---|---|---|---|
| 1 | **User Input** | Allows the user to enter story prompt, character name, setting, story tone and art style. | `index.html` / FastAPI | **Done** | High |
| 2 | **AI Story Generation** | Generates a structured comic story from the user's inputs using Gemini AI. | `gemini_flash.py` / `routes.py` | **Done** | High |
| 3 | **Five-Panel Comic Creation** | Converts the generated story into five individual comic panels. | `routes.py` / `layout_builder.py` | **Done** | High |
| 4 | **AI Image Generation** | Generates illustrations for the comic panels using Hugging Face AI image generation. | `image_generator.py` | **Done** | High |
| 5 | **Comic Preview** | Displays the generated comic panels with story content in the browser. | `comic_preview.html` / CSS | **Done** | High |
| 6 | **PDF Export** | Creates a downloadable PDF containing the generated comic. | `exporters.py` / `routes.py` | **Done** | Medium |
| 7 | **API Documentation** | Provides interactive API documentation for testing ComicCraft endpoints. | FastAPI `/docs` | **Done** | Medium |

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| **8** | **Error Handling** | Handles API and image-generation errors and provides appropriate responses. | `routes.py` / `image_generator.py` | **Done** | Medium |

**Feature Category Breakdown:**

| S. No | Category | Features in Category | Example Features |
|---|---|---|---|
| 1 | User Interface (UI) | 2 | User Input, Comic Preview |
| 2 | Backend / Logic | 3 | Story Generation, Five-Panel Creation, Error |
| 3 | Database / Storage | 1 | Generated comic images and PDF files stored locally |
| 4 | API / Integration | 2 | Gemini AI Integration, Hugging Face Image Generation |
| 5 | Security / Authentication | 3 | API keys stored securely using `.env` configuration |

## *Feature Summary*

| Metric | COUNT / VALUE |
|---|---|
| **TOTAL FEATURES PLANNED** | **8** |
| **TOTAL FEATURES IMPLEMENTED** | **8** |
| **CORE / MUST-HAVE FEATURES** | **5** |
| **ADDITIONAL / NICE-TO-HAVE FEATURES** | **3** |
| **FEATURES TESTED & VERIFIED** | **8** |